

**Q1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.**

**Code:**

```
interface Shape {
    void area();
}

class Circle implements Shape {
    double radius = 5;

    public void area() {
        System.out.println("Circle Area = " + (3.14 * radius * radius));
    }
}

class Rectangle extends Circle {
    double length = 10;
    double width = 5;

    public void area() {
        System.out.println("Rectangle Area = " + (length * width));
    }
}

public class ShapeDemo {
    public static void main(String[] args) {

        Circle c = new Circle();
        Rectangle r = new Rectangle();

        c.area();
        r.area();
    }
}
```

**OUTPUT:**

```
Circle Area = 78.5
Rectangle Area = 50.0
```

**Q2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.**

**Code:**

```
interface Product {  
    void display();  
}
```

```
class ProductDetails {  
    String name;  
    double price;  
  
    ProductDetails(String name, double price) {  
        this.name = name;  
        this.price = price;  
    }  
}
```

```
class Electronics extends ProductDetails implements Product {  
  
    Electronics(String name, double price) {  
        super(name, price);  
    }  
  
    public void display() {  
        System.out.println("Electronic Product: " + name);  
        System.out.println("Price: " + price);  
    }  
}
```

```
class Clothing extends ProductDetails implements Product {  
  
    Clothing(String name, double price) {  
        super(name, price);  
    }  
  
    public void display() {  
        System.out.println("Clothing Product: " + name);  
        System.out.println("Price: " + price);  
    }  
}
```

```
class Grocery extends ProductDetails implements Product {
```

```
    Grocery(String name, double price) {  
        super(name, price);  
    }  
}
```

```
    public void display() {  
        System.out.println("Grocery Product: " + name);  
        System.out.println("Price: " + price);  
    }  
}
```

```
public class ECommerce {  
    public static void main(String[] args) {  
  
        Electronics e = new Electronics("Laptop", 50000);  
        Clothing c = new Clothing("T-Shirt", 999);  
        Grocery g = new Grocery("Rice", 800);  
  
        e.display();  
        c.display();  
        g.display();  
    }  
}
```

## Output:

```
Electronic Product: Laptop  
Price: 50000.0  
Clothing Product: T-Shirt  
Price: 999.0  
Grocery Product: Rice  
Price: 800.0
```